

Warming of the Antarctic Sea Floor

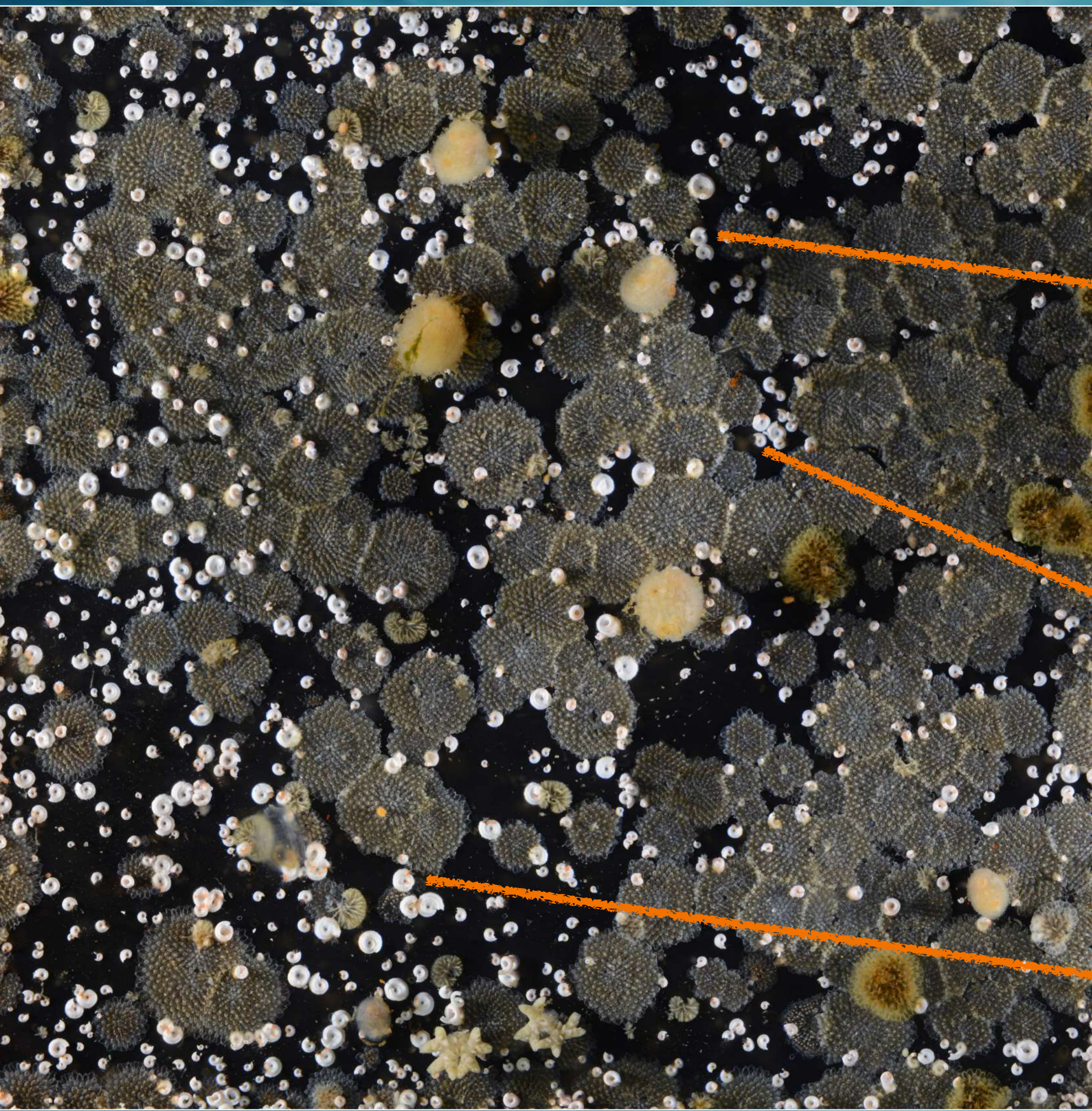
Determining responses to warming of Antarctic benthic communities using heated panels: An experimental approach

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Antarctic seas are forecast to warm: How will warming influence benthic communities through changes in recruitment, growth and community composition?

Warming drives significant changes to benthic communities in the Antarctic Peninsula (tube-worms as model organisms)



Responses of tube worms at +1°C and +2°C:

R. perrieri



Growth: +1° +2°
Assemblage proportion: +1° +2°

P. stalagmia



Growth: +1° +2°
Assemblage proportion: +1° +2°

P. levinseni

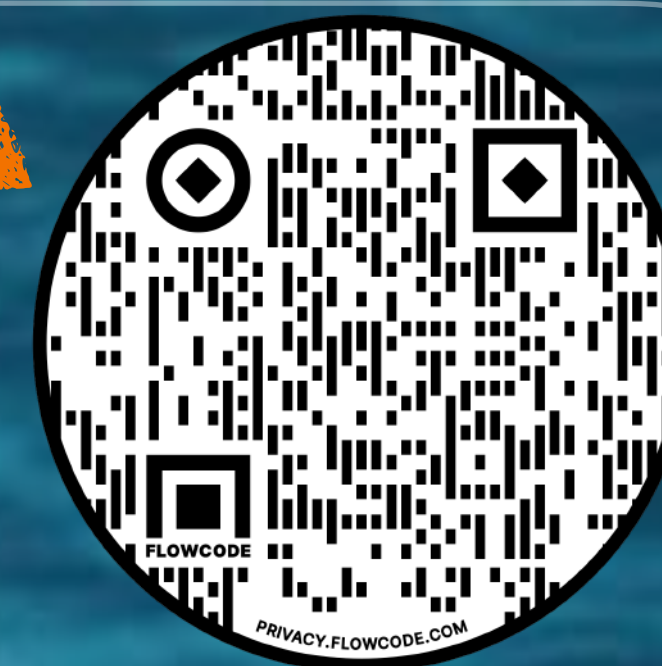


Growth: +1° +2°
Assemblage proportion: +1° +2°

Coastal seas of the Antarctic Peninsula are predicted to be +1°C and +2°C warmer over the next 50 and 100 years. Bespoke heated settlement panels allowed us to study responses to climate change at the Rothera Research Station (★ on the map), using tube-worms as an example of potential changes. These mesocosm experiments provided an indicator of how larger scale changes in community structure may be driven by species-specific responses to warming.

More information:

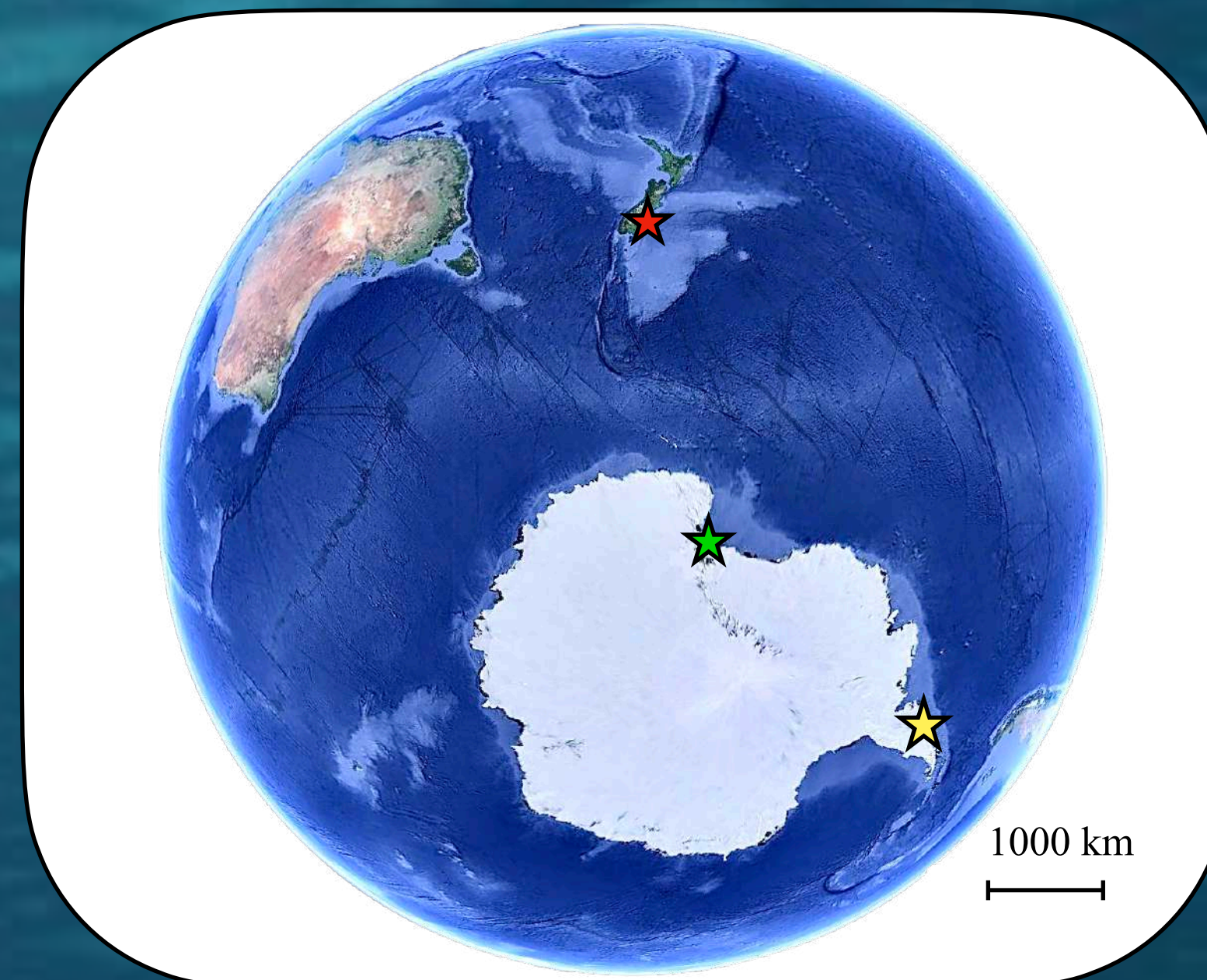
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FUTURE RESEARCH

Using novel heated settlement plates and working with the British Antarctic Survey (BAS) and the Antarctic Science Platform, we aim to examine warming effects on benthic communities in:

- ★ Otago Peninsula, New Zealand
- ★ The Ross Sea (McMurdo Sound and Terra Nova Bay), Antarctica



How do we heat the seafloor to look at community changes?



Shore-based power supply used to control the temperature

Cables

Metal heat trace in PVC panels located on concrete slabs on the benthos

Panels monitored *in situ* every month by SCUBA divers via digital photography



We use bespoke heated panels located on the seafloor (15 m depth) that are warmed through electrical cables and a controller on land. Panels are warmed up to +4°C above ambient over extended periods (months) and the responses are monitored by SCUBA diver photography. Changes are quantified using image analysis software of plate communities.

Photographs are merged to a single high resolution image, allowing to follow the assemblage and growth rate of species through time.

Quadrat sampling is used to analyse the community composition of species through months, while image analysis software is used to measure the growth rate of individuals per species and per panel through time.



Evolution of one panel and one tube-worm through time

Crop_014_17Mar15			
SPECIES	PROPORTION	%	
Perrieri	29	15.18	
Stalagmia	78	40.84	
Levinseni	84	43.98	
TOTAL	191	100	

GROWTH (mm)			
NUMBER	PERRIERI	STALAGMIA	LEVINSENI
1	1.083	0.832	1.069
2	1.256	1.146	0.782
3	1.180	0.876	0.967
4	1.069	0.882	1.056
5	1.070	1.027	0.994
6	1.065	0.898	1.233
7	1.198	0.962	1.169
8	1.165	0.880	1.082
9	1.470	0.920	1.109
10	1.248	0.907	1.118

Worksheet used for the quadrat sampling analysis

REFERENCES

- Ashton, G.V., Morley, S.A., Barnes, D.K.A., Clark, M.S. & Peck, L.S. (2017). Warming by 1°C Drives Species and Assemblage Level Responses in Antarctica's Marine Shallows. *Current Biology*, 27, 2698-2705.
- Clark, M.S., Nieva, L.V., Joseph I. Hoffman, J.I., Davies, A.J., Trivedi, U.H., Turner, F., Ashton, G.V. & Peck, L.S. (2019). Lack of long-term acclimation in Antarctic encrusting species suggests vulnerability to warming. *Nature communications*, 3383(10).